

A quick clarification

1. Going strictly by ITU's IMT-Advanced definition, only LTE-Advanced (Releases 10 and 11) and WiMAX 2.0 (IEEE 802.16m) can be considered as 4G services. These are not yet operational anywhere in the world. We can expect to see the first deployments only by the year-end or in 2013.
2. Countries like India, China, Japan and Brazil have deployed LTE (Releases 8 and 9) in recent months. Many players in the telecom industry tend to accept this as 4G as it uses multi-carrier OFDM technology and can be considered a major jump in technology and user/operator experience.

vast number of existing user devices in the marketplace, and inability to provide the high bar of voice reliability and quality. The VoIP alternative is primarily statistical in nature and lacks the advantages of traditional circuit-switched based voice, e.g., quality, deterministic behaviour in terms of latency and routing, and advanced supplementary services," says Dr Borkar.

Since 4G LTE has an all-IP core (enhanced packet core), it cannot directly support voice applications until voice-over-IP (VoIP) becomes the norm. Moreover, all operators have already invested heavily in 2G and 3G systems which use circuit-switching for voice support.

The general approach that current LTE operators are using is circuit switch fallback to a 2G or 3G network via the memory management entity in the enhanced packet core. This is transparent to the user device. In addition to the circuit switch fallback, the 4G user devices are designed to be multi-mode so that they can connect to a 2G or 3G system in case the LTE system is not present or the LTE signals are too weak.

"The 4G technology evolved and standardised by 3GPP has multiple options of supporting voice. One is circuit-switched voice. LTE supports circuit switch fallback to 3G/2G network for voice support. Advantages are definitely there for multi-service operators like Airtel, as it can offer varied services and bundle service plans to meet the requirements of the customers," says Rajiv Rajgopal, CEO-broadband/data, Bharti Airtel.

Looking into the future, David Maidment, mobile segment manager,

ARM, comments: "Moving forward, operators will want to break their dependency on circuit-switched services and move to an all-LTE deployment. As such, they are working to standardise and deploy voice-over-LTE (VoLTE), which allows voice to be carried over the packet network. VoLTE handsets are already available, and in the next one or two years we will begin to see the growth of these services."

Availability of 4G devices

True 4G equipment are not widely available today. LTE Release 8 systems have been deployed since 2010, but these are not 4G—merely 3.5G. As is expected in a family of standards like LTE, device and infrastructure manufacturers will provide migration and compatibility between different releases. Current LTE smartphones and other devices will operate in future LTE networks but may not be able to exploit the advanced and new features like carrier aggregation. Real 4G devices compatible with LTE-Advanced specifications may be available by the year-end.

"With the current technologies, many advanced features can be effected in the devices via provisioning and software updates," points out Dr Borkar.

"Initially, 4G user equipment targeted notebooks and tablets for data services. With the availability of chipsets that support both 3G and 4G technologies, the market is beginning to see 4G LTE smartphones," adds Matharu of Xilinx—a company that provides silicon devices, intellectual property solutions and design services for wireless infrastructure.

Rajgopal notes: "Bharti Airtel through its GTI, along with other partners like MCCI and Softbank, is taking a lead in ensuring that the device ecosystem rapidly matures to the operators' and consumers' requirements. Currently, we have an array of compatible devices like dongles and CPE devices that support TDD-based LTE. Smartphone and tablet users can access 4G LTE services over Wi-Fi networks, from CPE devices. We are working with the OEMs in ramping up the device ecosystem in the country."

That said, there is also the issue of devices made for one region not working in others. To this, Rajgopal responds, "At present, 4G LTE devices available in the open market or USA are not compatible in India. We are working with OEMs and chipset vendors to ensure that this will not be an impediment. As a matter of fact, Bharti Airtel is one of the founding members of the GTI that is working on standards for TD LTE."

Roaming issues

Frequency bands and duplex operation are the two primary reasons why equipment used in one continent are not compatible in others.

"LTE as a standard can be deployed in over 40 different frequency bands. This is a benefit as it makes the standard flexible, but it's also a weakness from a fragmentation and roaming point of view. We have already seen issues around band support in LTE with current-generation devices that are unable to roam outside their home networks. Multi-band support is nothing new in cellular applications. From 2G onwards equipment vendors have been producing multi-band phones. The issue with LTE has been the sheer number of bands as well as the fact that it supports two duplexing methods—FDD and TDD," explains Maidment.

"From a signal processing point of view it is already possible to implement a 'world phone' that can span FDD and TDD. The issue is in the